Using Slope-Intercept Form

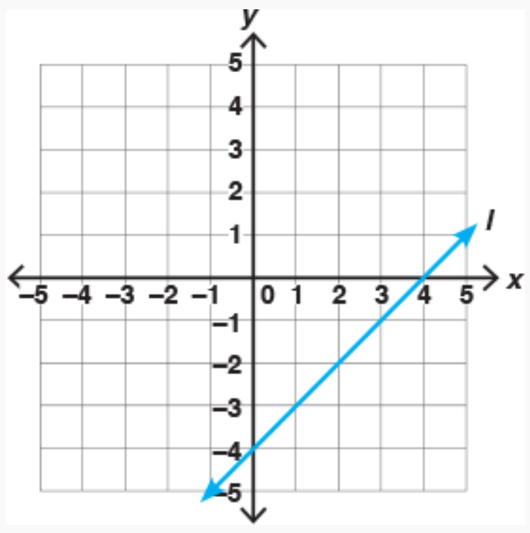
By looking at a line or its points, you can write the equation that corresponds with the line. All you need are the slope of the line and the *y*-intercept.

The **slope-intercept** form of an equation is $y = mx + b$, where m is the slope and b is the y-value of the y-intercept. Find the slope and the y-intercept, and then substitute those values into the slope-intercept equation.

Example 1

Hide

Write the equation of line *l*.



Step 1

Pick two points such as (3, −1) as (x_1, y_1) and (2, −2) as (x_2, y_2) and find the slope.

$m = \frac{-2-(-1)}{2-3} = \frac{-1}{-1} = 1$

Step 2

Find the y-intercept.

The line crosses the y-axis at (0, −4).

$b = -4$

Step 3

Substitute both values into the equation.

$y = mx + b$

$y = 1x + (-4)$

$y = x - 4$

The equation of the line is **y = x − 4.**

You can use slope-intercept form to graph an equation. Plot the y-intercept on the graph. Then use the slope to count rise and run.

Example 2

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Graph the line $y = \frac{1}{2}x - 3$

Step 1

Identify the slope and the y-intercept: $m = \frac{1}{2}$ and $b = -3$

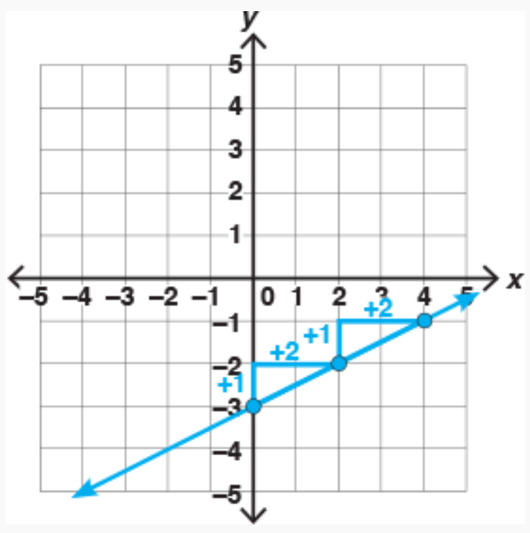
Step 2

Plot the y-intercept at (0, −3).

Step 3

Count out the slope.

The line rises 1 and runs 2. Count up 1 unit and to the right 2 units and plot a point. Good test-takers plot 3 points before drawing the line.



Key Terms

- **point-slope form**
- **standard form**
- **slope-intercept form**

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